

# PE Numericals Unit I

## Tut 3 Q 1

A thyristor can be triggered if  $\frac{dv}{dt} = 190 \text{ V}/\mu\text{s}$ .

If the capacitive current flowing through the junction is 8 mA, find the equivalent capacitance of depletion layer.



## Tut 4 Q 1

A string of four series connected thyristors is provided with static & dynamic equalizing circuits. This string has to withstand an off state voltage of 10 KV. The static equalizing resistance is  $25000\ \Omega$  & the dynamic equalizing circuit has  $R_c = 40\ \Omega$  &  $C = 0.08\ \mu\text{F}$ . The leakage currents for four thyristors are 21 mA, 25 mA, 18 mA & 16 mA respectively. Determine voltage across each SCR in the off state & the discharge current of each capacitor at the time of turn on.



## Tut 4 Q 3

It is required to operate 250 A SCR in parallel with 350 A SCR with their respective on state voltage drops of 1.6 V & 1.2 V. Calculate the value of resistance to be inserted in series with each SCR so that they share the total load of 600 A in proportion to their current ratings.

## Tut 4 Q 4

Q.4.) SCRs with a rating of 1000 V & 200 A are available to be used in a string to handle 6 KV & 1 KA. Calculate the number of series & parallel units required incase derating factor is

(a) 0.1 (b) 0.2